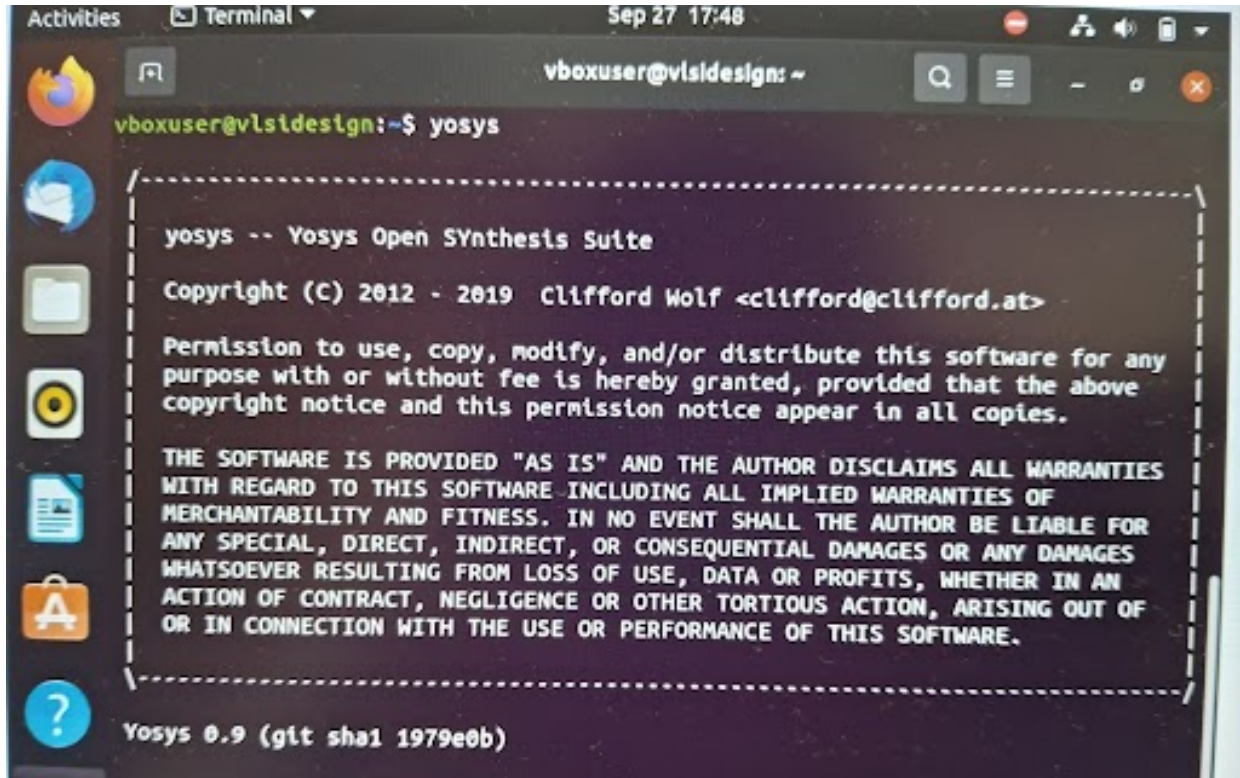


If sudo authentication fails for vboxuser, use the following steps:

```
$ sudo apt update
```

```
$ sudo apt install git
```

A screenshot of a Linux terminal window. The window title is "Terminal" and the date/time is "Sep 27 17:48". The user is "vboxuser@vlsdesign: ~". The command "yosys" has been executed. The output is enclosed in a dashed box and includes the Yosys Open SYNthesis Suite version information, copyright notice, and a disclaimer. Below the box, it shows "Yosys 0.9 (git sha1 1979e0b)".

```
vboxuser@vlsdesign:~$ yosys

yosys -- Yosys Open SYNthesis Suite

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ACTION OF CONTRACT, NEGLIGENCE OR OTHER TORTIOUS ACTION, ARISING OUT OF
OR IN CONNECTION WITH THE USE OR PERFORMANCE OF THIS SOFTWARE.

Yosys 0.9 (git sha1 1979e0b)
```



yosys> exit



End of script. Logfile hash: 0701883ad7
CPU: user 0.03s system 0.01s, MEM: 20.80 MB total, 9.70 MB resident
Yosys 0.9 (git sha1 1979e0b)
Time spent: 100% 1x cd (0 sec)



vboxuser@vlsidesign:~\$ sudo apt-get install iverilog

[sudo] password for vboxuser:

Reading package lists... Done

Building dependency tree

Reading state information... Done

iverilog is already the newest version (10.3-1build1).

0 upgraded, 0 newly installed, 0 to remove and 365 not upgraded.

vboxuser@vlsidesign:~\$ iverilog

iverilog: no source files.



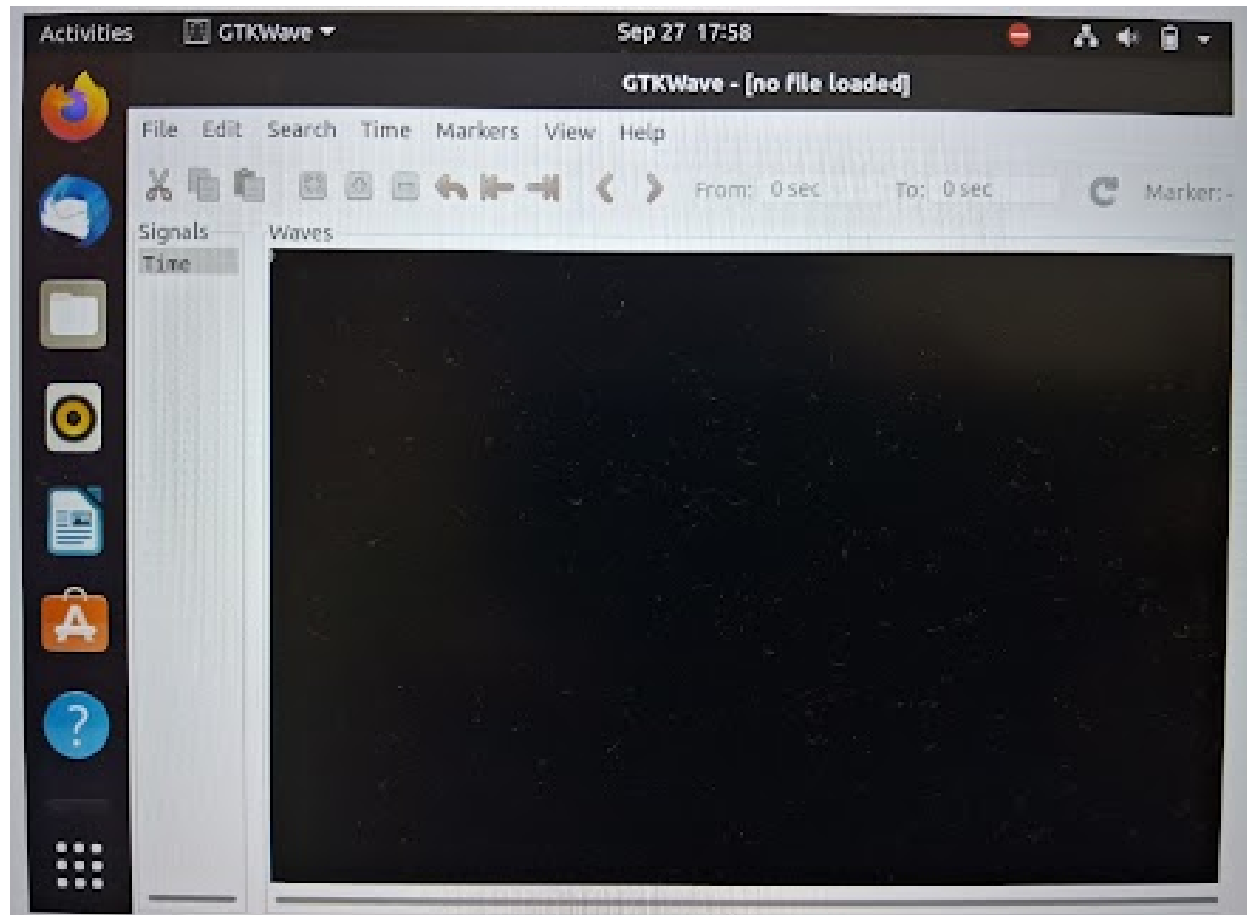
Usage: iverilog [-ESvV] [-B base] [-c cndfile|-f cndfile]
[-g1995|-g2001|-g2005|-g2005-sv|-g2009|-g2012] [-g<feature>]
[-D macro[=defn]] [-I includedir]
[-M [mode=]depfile] [-n module]
[-N file] [-o filename] [-p flag=value]
[-s topmodule] [-t target] [-T min|typ|max]
[-W class] [-y dir] [-Y suf] [-l file] source_file(s)



See the man page for details.

vboxuser@vlsidesign:~\$

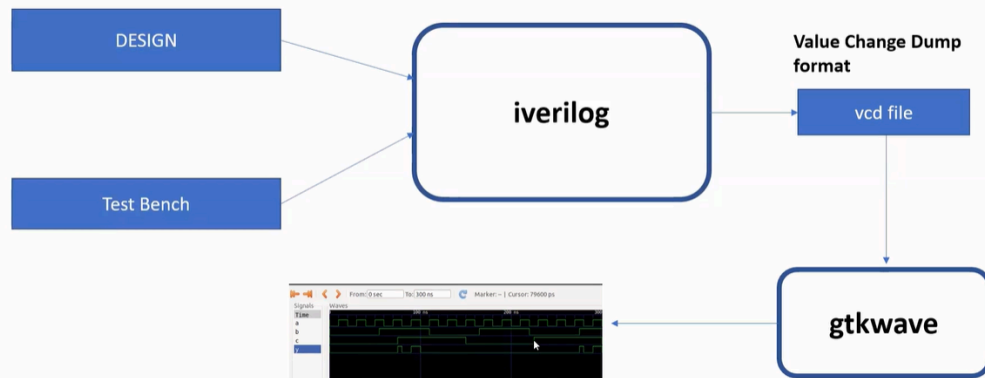




Day 1:

```
$ git clone  
https://github.com/kunalg123/sky130RTLDesignAndSynthesisWorkshop.git  
$ ls
```

Iverilog based Simulation Flow



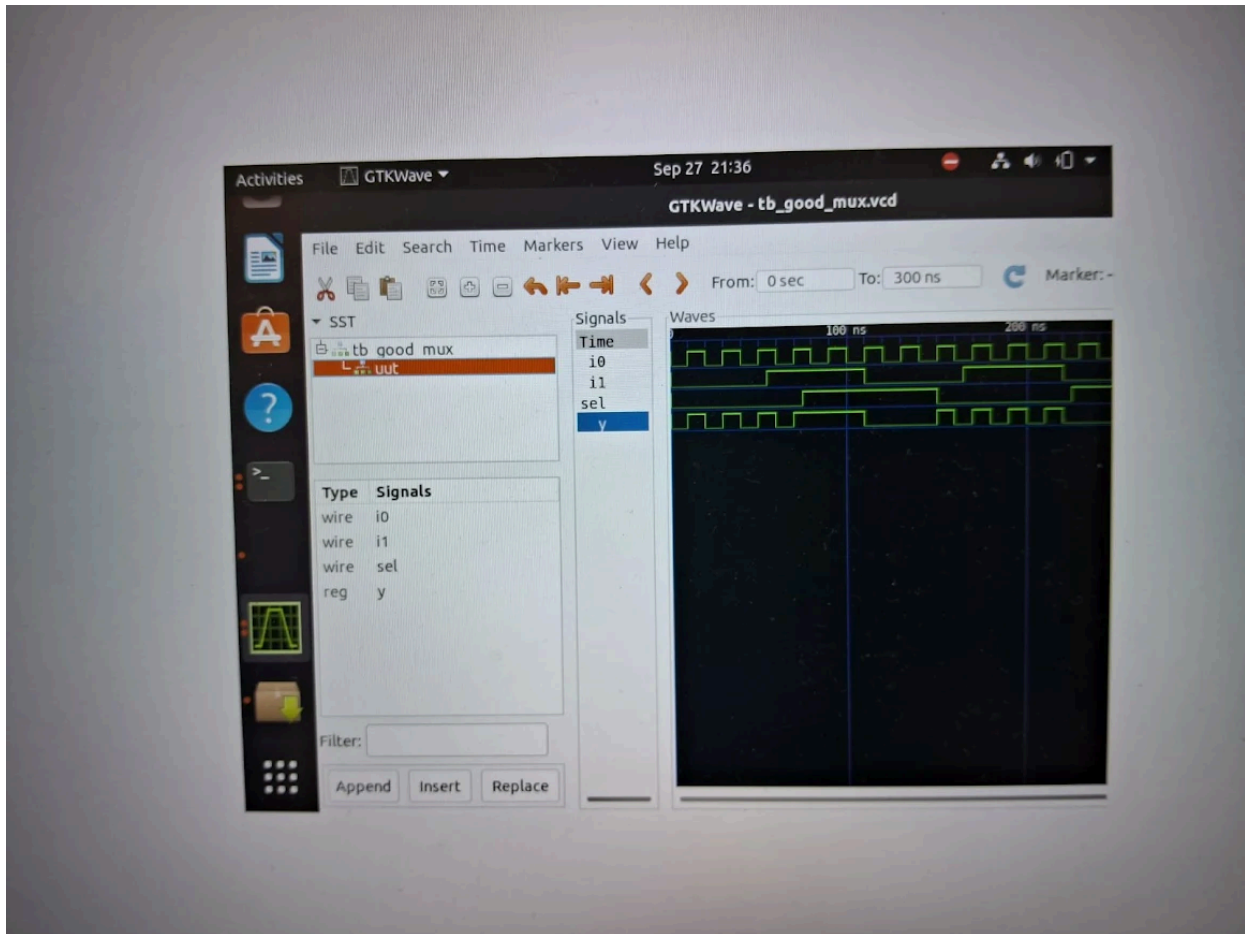
```
$ iverilog good_mux.v tb_good_mux.v
```

```
$ ls
```

```
$ ./a.out
```

```
(VCD info:dumpfile tb_good_mux.vcd opened for output)
```

```
$ gtkwave tb_good_mux.vcd
```



```
$ sudo apt install vim  
$ sudo apt install -y vim-gtk3  
$ givm tb_good_mux.v -o good_mux.v
```

ties GVim Sep 27 21:53

vboxuser@vlsidesign: ~/sky130RTLDesignAndSynthesisWor...

tb_good_mux.v (-/sky130RTLDesignAndSynthesisWor.../verilog_files) (1 of 2) - GVIM

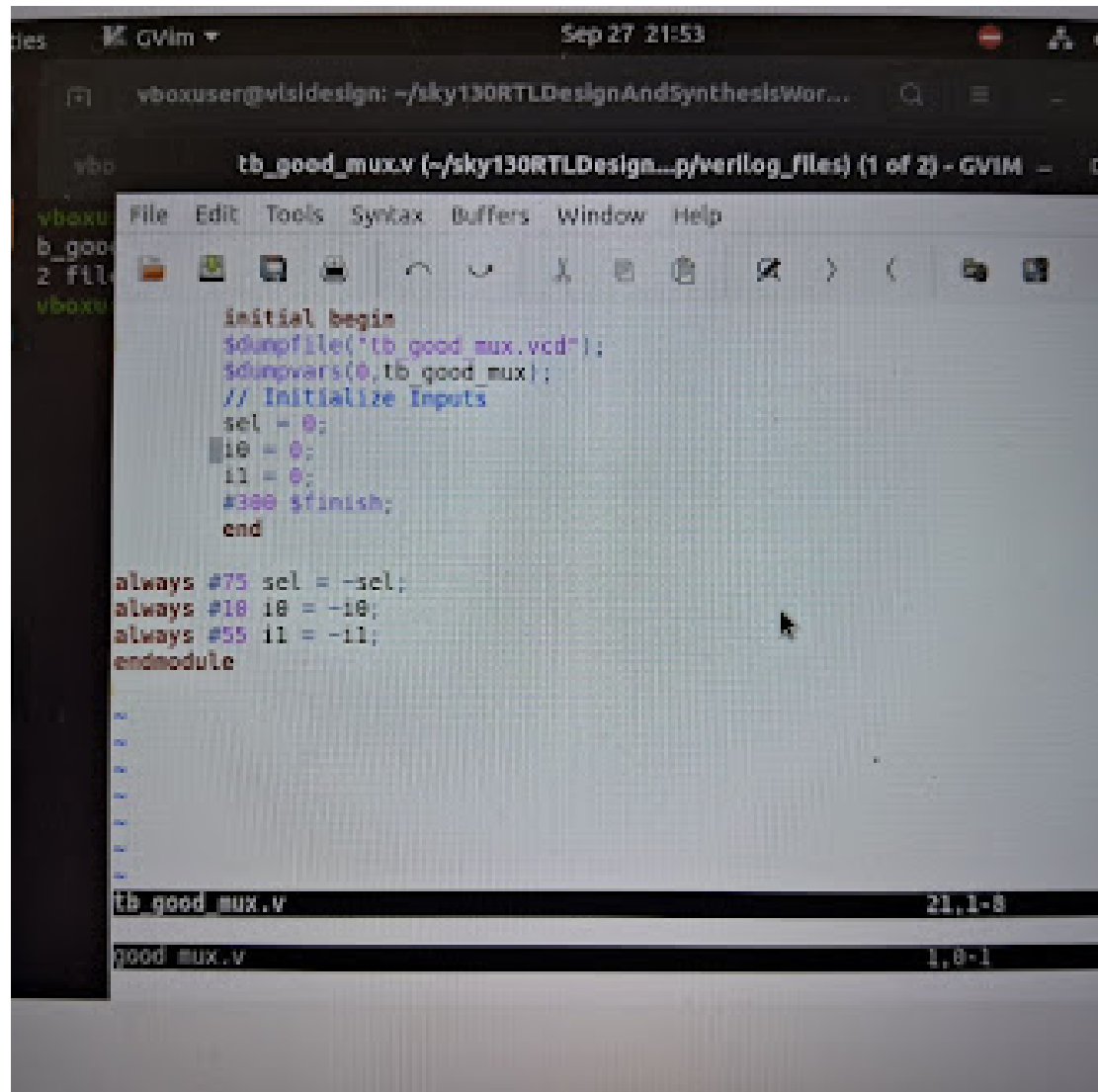
File Edit Tools Syntax Buffers Window Help

```
timescale 1ns / 1ps
module tb_good_mux;
    // Inputs
    reg i0,i1,sel;
    // Outputs
    wire y;

    // Instantiate the Unit Under Test (UUT)
    good_mux uut (
        .sel(sel),
        .i0(i0),
        .i1(i1),
        .y(y)
    );

    initial begin
        $dumpfile("tb_good_mux.vcd");
        $dumpvars(0,tb_good_mux);
        // Initialize Inputs
        sel = 0;
        i0 = 0;
        i1 = 0;
    end
endmodule
```

tb_good_mux.v	9,1-8
good_mux.v	1,0-1



nvim

gvim

Sep 27 21:53

vbouser@vlsidesign: ~/sky130RTLDesignAndSynthesisWor...

tb_good_mux.v (~/.sky130RTLDesign.../verilog_files) (1 of 2) - GVIM

File Edit Tools Syntax Buffers Window Help

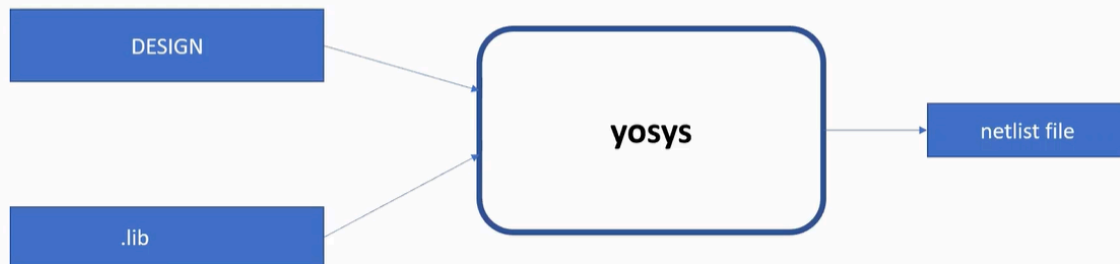
tb_good_mux.v 30,0-1

module good_mux (input i8 , input i1 , input sel , output reg y);
always @ (*)
begin
 if(sel)
 y <= i1;
 else
 y <= i8;
end
endmodule

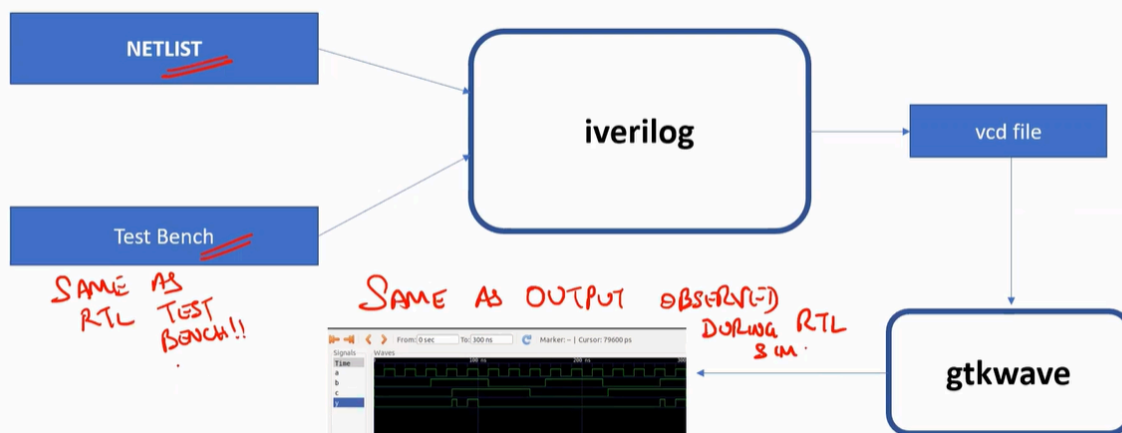
good_mux.v 1,0-1

Synthesizer

- Tool used for converting the RTL to netlist
- **Yosys** is the synthesizer used in this course



Verify the synthesis



NOTE

The set of Primary inputs / primary outputs will remain same between the RTL design and Synthesized netlist → Same Test bench can be used !!

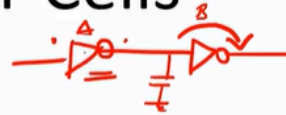
A **Netlist** is a representation of the digital design in the form of a list of interconnected **standard cells** from a specific technology library, where the functional and timing characteristics of those cells are defined in the associated **.lib (Liberty) files**.

RTL: behavioural representation of the specification

RTL design is converted to gates and connections are made between them resulting in a Netlist file, .lib contains standard forms of all these types of standard cells such as AND, OR etc.

Combinational delay in logic path determines the maximum speed of operation of the digital logic circuit. Why different low and medium cells in .lib then? - To avoid “HOLD” issues at the capture flop!

Faster Cells vs Slower Cells



- Load in Digital Logic circuit → Capacitance
- Faster the charging / discharging of capacitance → Lesser the cell delay
 - To charge / discharge the capacitance fast , we need transistors capable of sourcing more current → **WIDE TRANSISTORS**
 - Wider transistors -> Low Delay -> More Area and Power as well !!
 - Narrow transistors -> More Delay -> Less Area and Power
 - Faster cells donot come free , they come at penalty of area and power

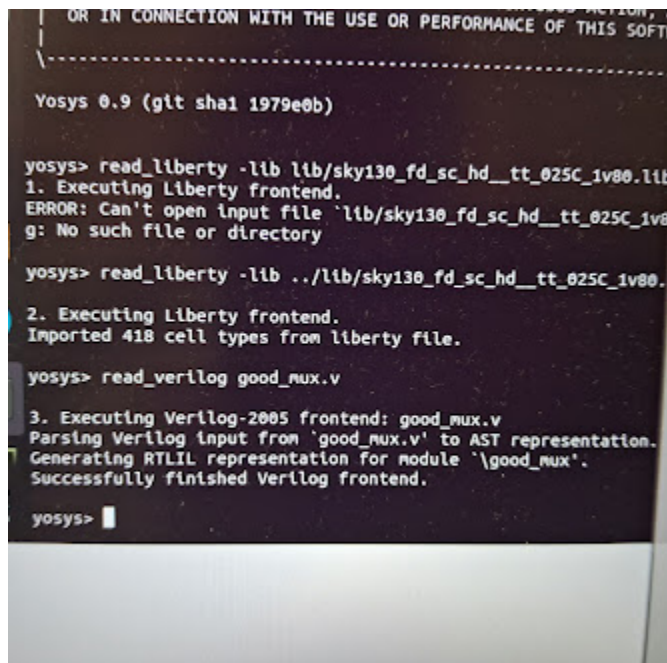
Selection of Cells

- Need to guide the Synthesizer to select the flavour of cells that is optimum for the implementation of logic circuit
- More use of faster cells
 - Bad circuit interms of Power and Area
 - Hold time violations ??
- More use of slower cells
 - Sluggish circuit , may not meet the performance need

Constraints is the file wherein we define which cells to use and how exactly.

```
$ yosys
```

```
$ read_liberty -lib ../lib/sky130_fd_sc_hd__tt_025C_1v80.lib
```



```
OR IN CONNECTION WITH THE USE OR PERFORMANCE OF THIS SOFTWARE.
-----
Yosys 0.9 (git sha1 1979e0b)

yosys> read_liberty -lib lib/sky130_fd_sc_hd__tt_025C_1v80.lib
1. Executing Liberty frontend.
ERROR: Can't open input file 'lib/sky130_fd_sc_hd__tt_025C_1v80.lib': No such file or directory

yosys> read_liberty -lib ../lib/sky130_fd_sc_hd__tt_025C_1v80.lib
2. Executing Liberty frontend.
Imported 418 cell types from liberty file.

yosys> read_verilog good_mux.v
3. Executing Verilog-2005 frontend: good_mux.v
Parsing Verilog input from 'good_mux.v' to AST representation.
Generating RTLIL representation for module '\good_mux'.
Successfully finished Verilog frontend.

yosys> 
```

```
$ synth -top good_mux
```

```

vboxuser@visdesign: ~/sky130RTLDesignAndSynthesisWor...
vboxuser@visdesign: ~/sky130RTL... x vboxuser@visdesign: ~/sky

4.25.1. Analyzing design hierarchy..
Top module: \good_mux

4.25.2. Analyzing design hierarchy..
Top module: \good_mux
Removed 0 unused modules.

4.26. Printing statistics.

=== good_mux ===
Number of wires:          4
Number of wire bits:      4
Number of public wires:   4
Number of public wire bits: 4
Number of memories:       0
Number of memory bits:    0
Number of processes:      0
Number of cells:          1
$MUX_                     1

4.27. Executing CHECK pass (checking for obvious problems).
checking module good_mux..
found and reported 0 problems.

yosys>

```

\$(#to generate netlist) abc -liberty ../lib/sky130_fd_sc_hd__tt_025C_1v80.lib

```

vboxuser@visdesign: ~/sky130RTLDesignAndSynthesisWor...
vboxuser@visdesign: ~/sky130RTL... x vboxuser@visdesign: ~/sky130RTL...

= 0.09 sec
ABC: Memory = 15.77 MB, Time = 0.09 sec
ABC: Warning: Detected 9 multi-output gates (for example, "sky130_fd_sc_hd__mux2_1").
ABC: + strash
ABC: + ifraig
ABC: + scorrr
ABC: Warning: The network is combinational (run "fraig" or "fraig_sweep").
ABC: + dc2
ABC: + dretine
ABC: + retine
A UbuntuSoftware
ABC: + asch -f
ABC: + knf
ABC: + kput
ABC: + write_blif <abc-temp-dir>/output.blif

5.1.2. Re-integrating ABC results.
ABC RESULTS: sky130_fd_sc_hd__mux2_1 cells: 1
ABC RESULTS: internal signals: 0
ABC RESULTS: input signals: 3
ABC RESULTS: output signals: 1
Removing temp directory.

yosys>

```

\$ show

